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August 6, 2025

Schematic Review 8/1/25

Chip Odyssey

Full Chain and System Specs

Later you propose a 6 bit ADC. That means that the min current to be measured is about $150\mu/2^6$ or $2.3\mu\text{A}$

What is a good electrical model for your input signal source?

Will you DC couple it?

$1\text{n} \times 1\text{K} = 1\mu\text{V}$ and $150\mu \times 1\text{K} = 150\text{mV}$
What does 250mA mean?

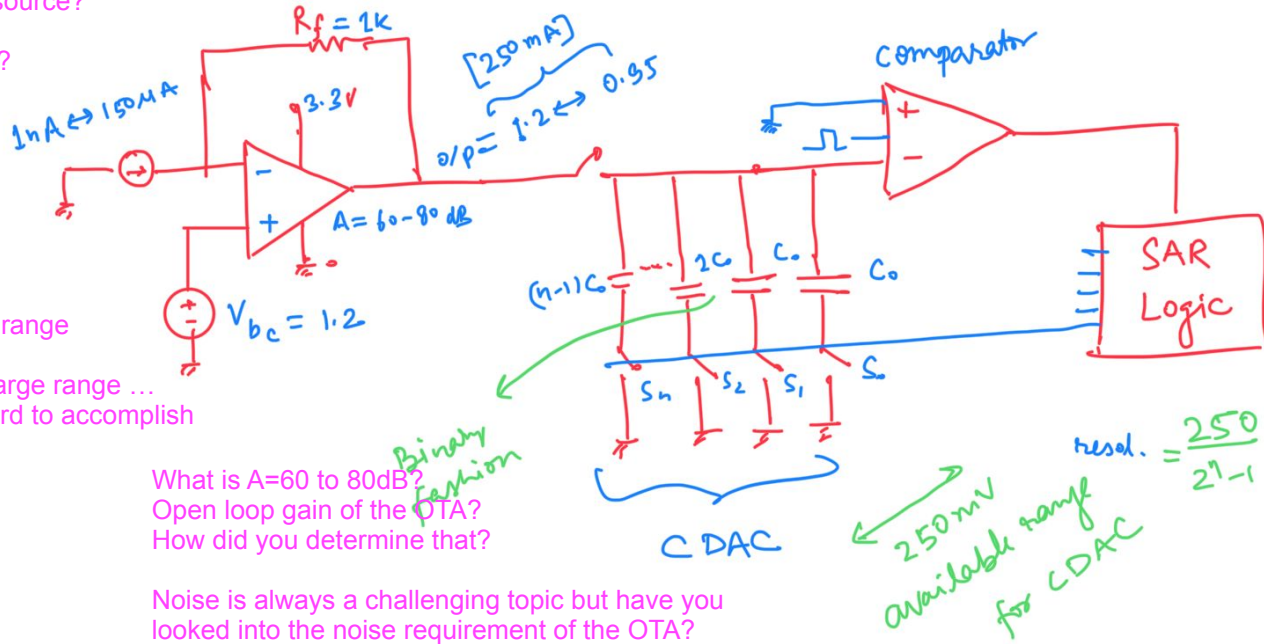
ADCs are typically designed with as large an input range as possible. So you should consider additional amplification between the TIA and ADC. Otherwise you will have to spend more 'effort' in the comparator design (low offset ...)

$150\mu/1\text{n}$ is 150,000
or 2^{17} or 17 bits
or >100dB dynamic range

That is a very, very large range ...
which will be very hard to accomplish

What is $A=60$ to 80dB ?
Open loop gain of the OTA?
How did you determine that?

Noise is always a challenging topic but have you looked into the noise requirement of the OTA?

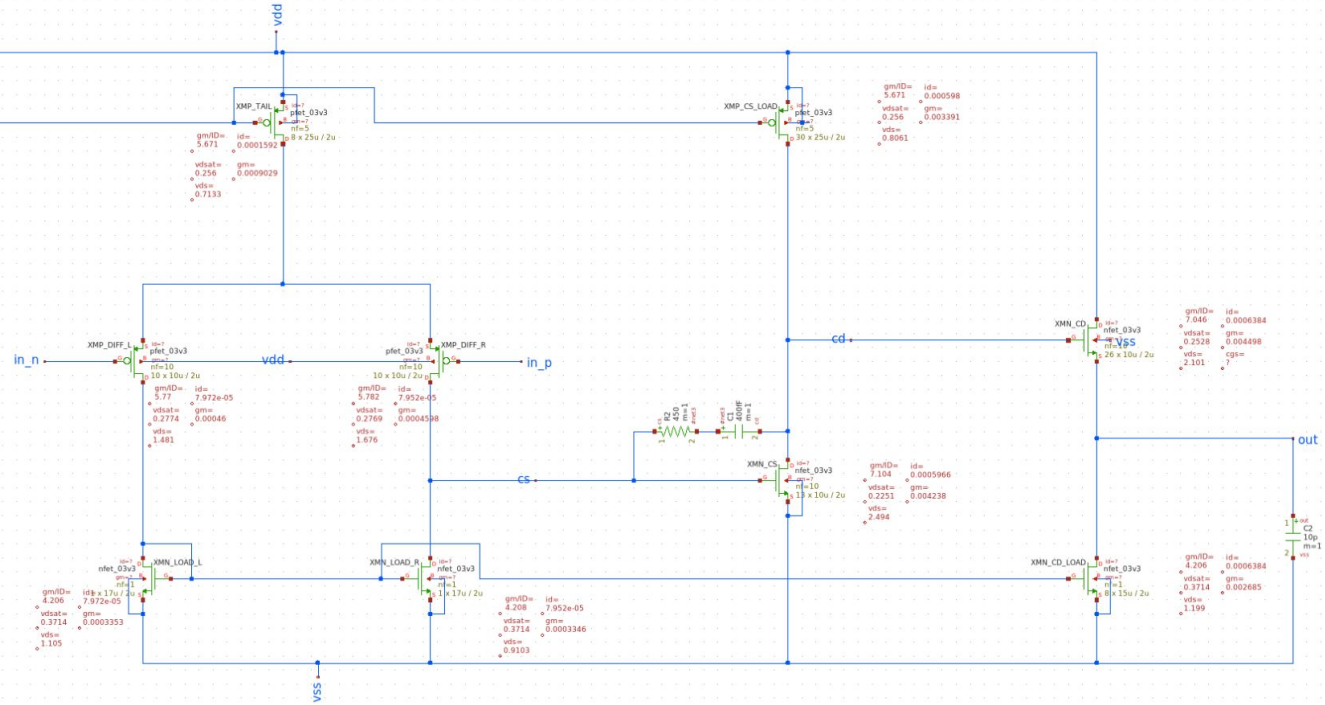


TIA

$I_1 = 160 \text{ uA}$

$I_2 = 600 \text{ uA}$

$I_3 = 600 \text{ uA}$



TIA

Sizing:

Using the gm/Id methodology with a custom VSCode sizing script.

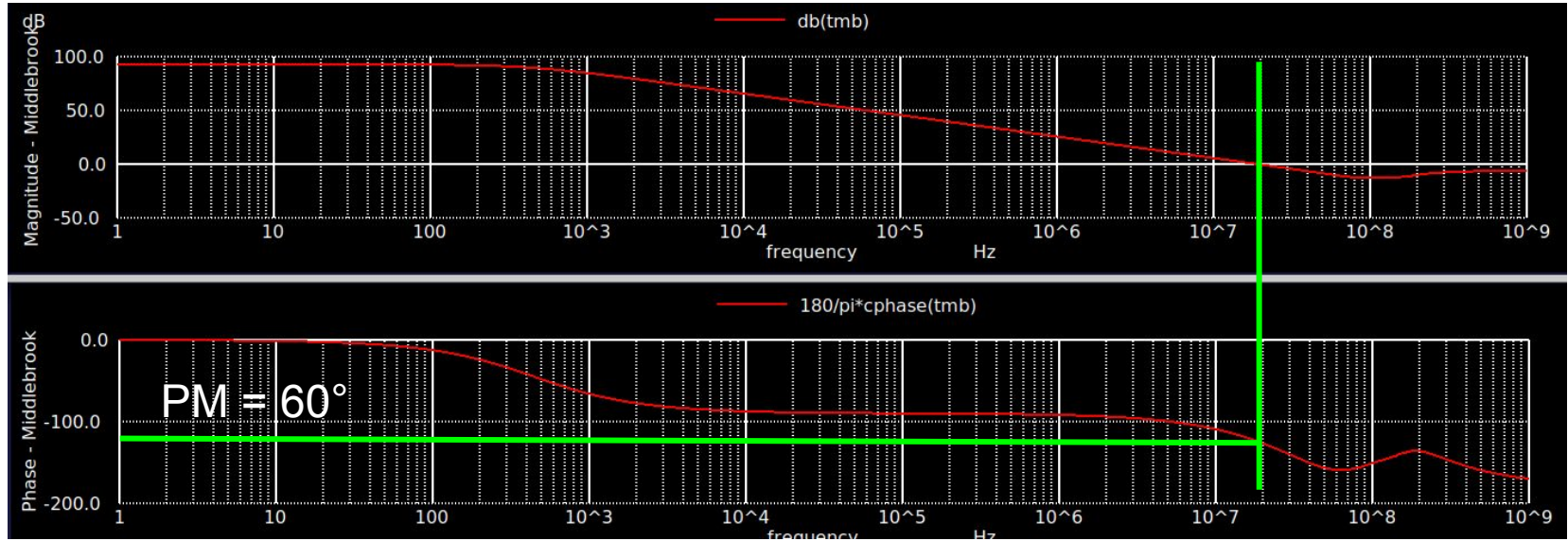
Design for a small gm/Id ≈ 5 to save area due to the high currents.

Given the high sensitivity you are looking for and thus low noise, large gm/I might serve you better

What are you showing? Loop Gain? What is the circuit configuration?

TIA: Stability Simulation

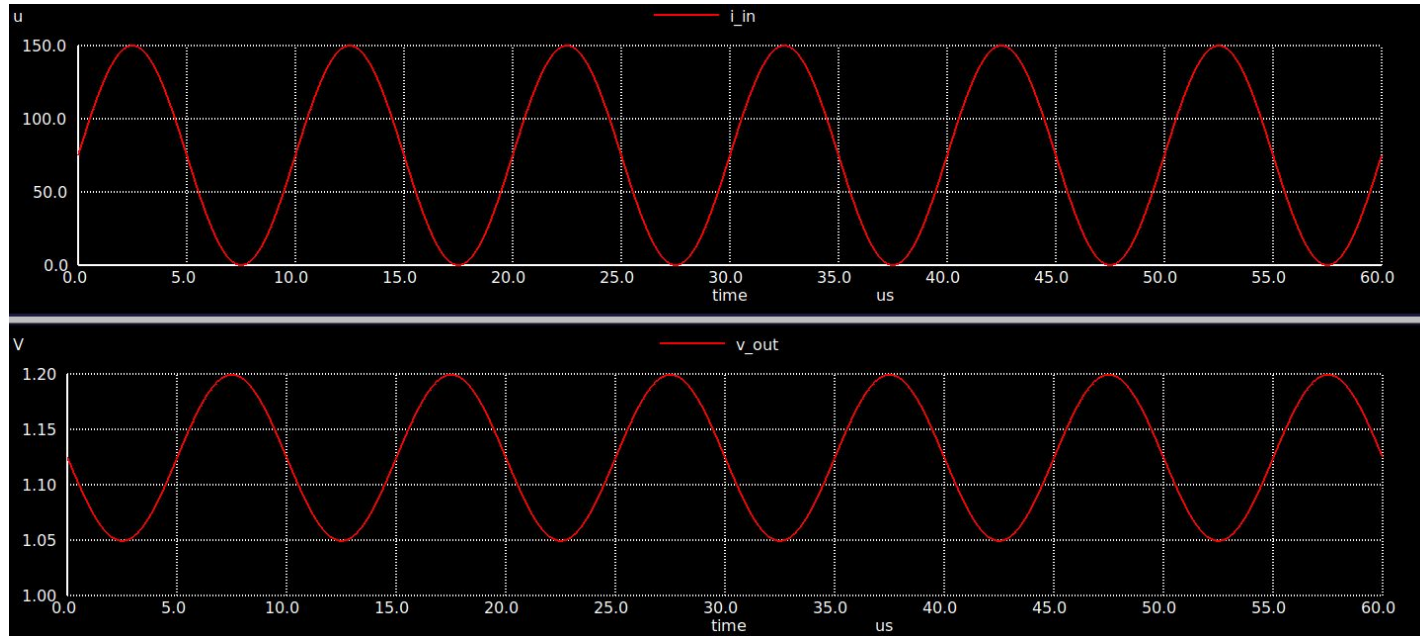
Remark: Stability simulation carried out with Middlebrook methode



TIA: Transient Simulation

Sim. schematic? What kind of input signal source model are you using? Make sure to use a practical source model, not an ideal current source. At least limit the output impedance.

Remark: $f = 100 \text{ kHz}$; $I_{\text{range}} = 1 \text{ nA} - 150 \text{ uA}$



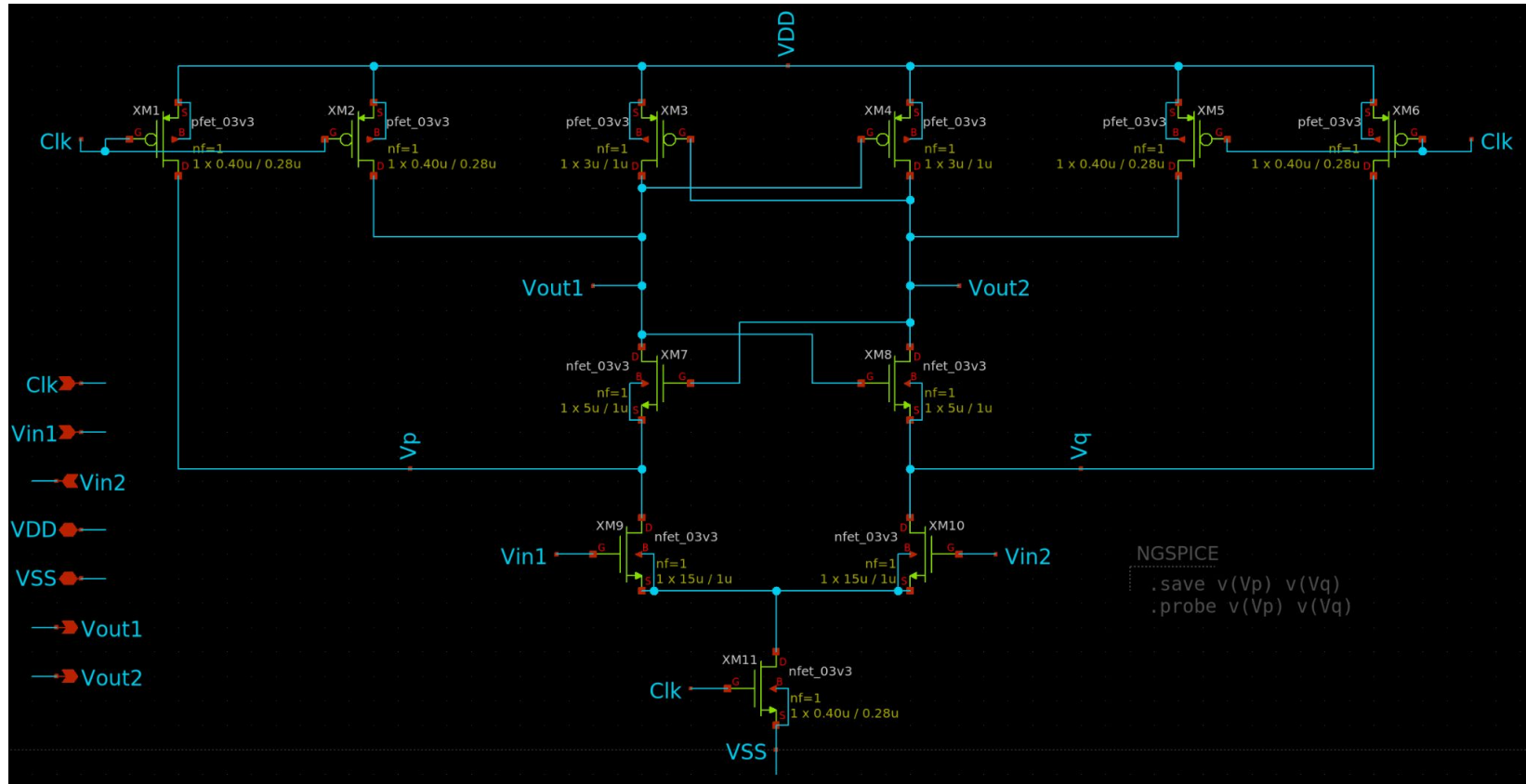
TIA: Linearity

Comparator Specs

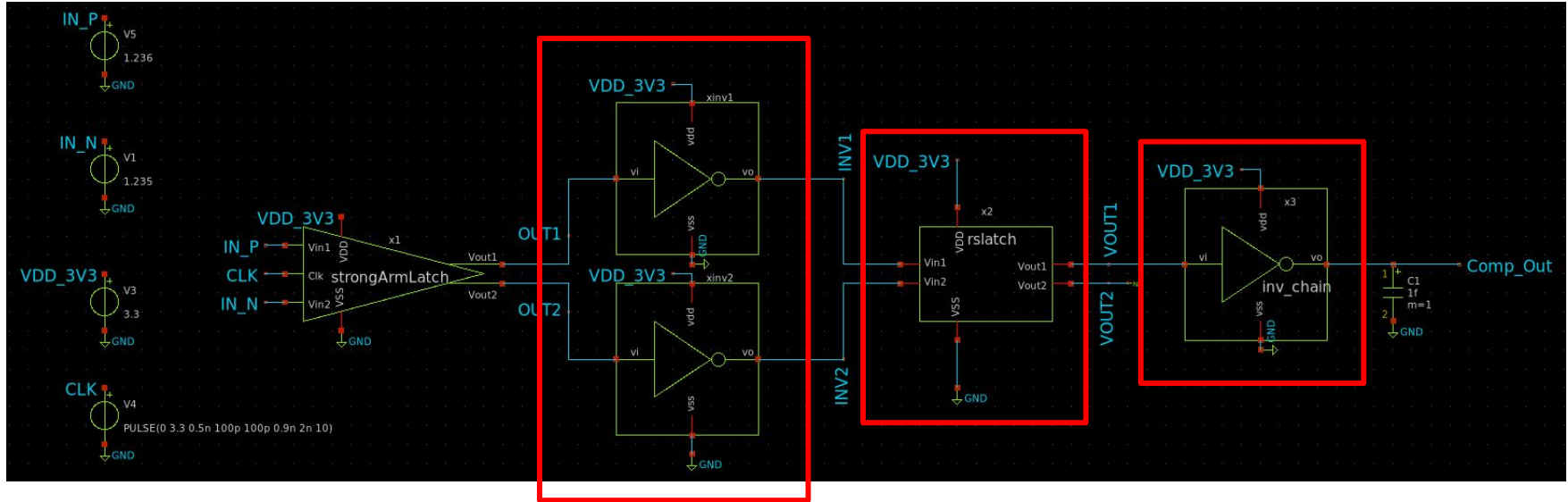
<u>Input Offset</u>	<u>Clock Freq</u>	<u>Power</u>	<u>Input Referred Noise</u>
< 2.34mV	> 1.5 MHz	< 2mW	< 1mV

- We have an input AC signal from 0.9V-1.2V so max input swing is 0.3V
 - Chose a 6-bit ADC so our LSB = $0.3 / 2^6 = 4.68\text{mV}$
 - Typically want our input offset to be less than $\frac{1}{2}$ of the LSB so < 2.34mV
- We chose to set our ADC sampling rate at 250kHz and for a 6-bit SAR ADC, it takes 6 cycles to determine the final value so our comparator and SAR logic have to work at a clock frequency of $> 250\text{kHz} * 6 = 1.5\text{MHz}$
- Ideally we want it to be somewhat low power and just set a reasonable spec of 2mW
- We don't want too much input referred noise or it will affect our ADC ENOB and it will get too close to the LSB value

Strong Arm Latch Comparator Schematic



Comparator Basic Operation Testbench Schematic



Output buffer to regenerate latch outputs

RS latch to hold output states

Inverter chain from 1x to 4x sizing to drive high impedance output capacitance

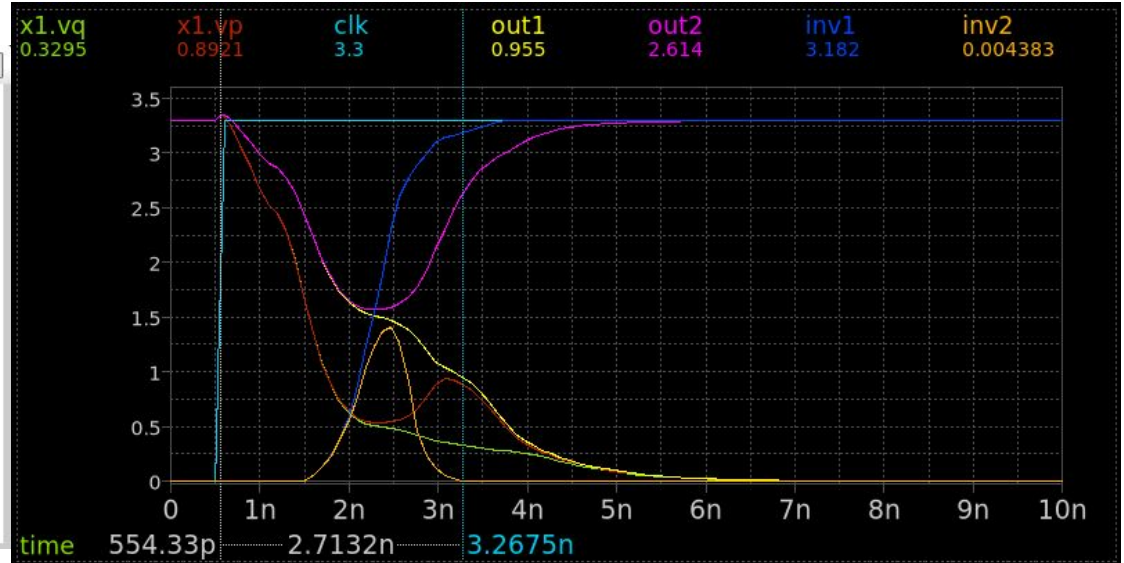
Slow Motion Operation (1mV input difference)

```
tb_fixed.spice" -a || sh
No. of Data Rows : 2527
Measurements for Transient Analysis

Error: measure t_vxvy_1650mw find(AT) : out of interval
.measure tran t_vxvy_1650mw find time when v(vdiff)=1.65 cross=1 failed!

y_vp      = 3.300000e+00
y_vq      = 3.300000e+00
vdyn_offset = 5.04144e-08
t_clk_vdd2  = 5.500000e-10
t_vxvy_n1650mw = 3.262651e-09
Netlist line no. 41:
Undefined parameter [t_vxvy_1650mw]
Netlist line no. 41:
Expression err: t_vxvy_1650mw-t_clk_vdd2}
Netlist line no. 41:
Cannot compute substitute
comp_speed = failed
comp_nspped = 1.84322e+08

binary raw file "tb_fixed.raw"
ngspice 1 -> █
```

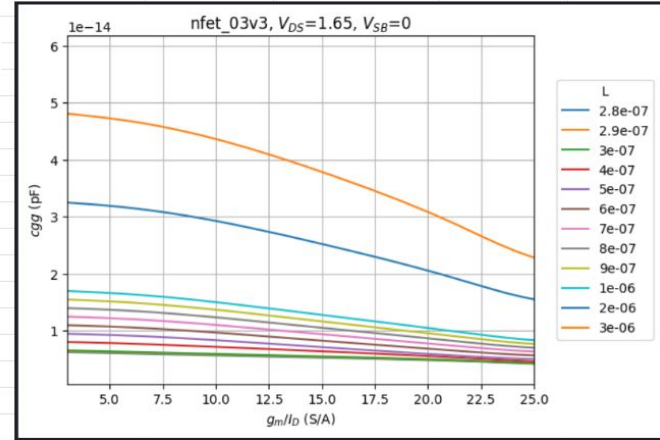


- Dynamic offset at input transistor drain nodes before next clock cycle is 50nV at 5MHz clock
- Comparator max speed assuming low output capacitance (100f) is 184MHz

Input Offset Voltage

The mismatch formula will give you a sigma (std. dev) of the offset distribution.
For how many sigma are you designing?

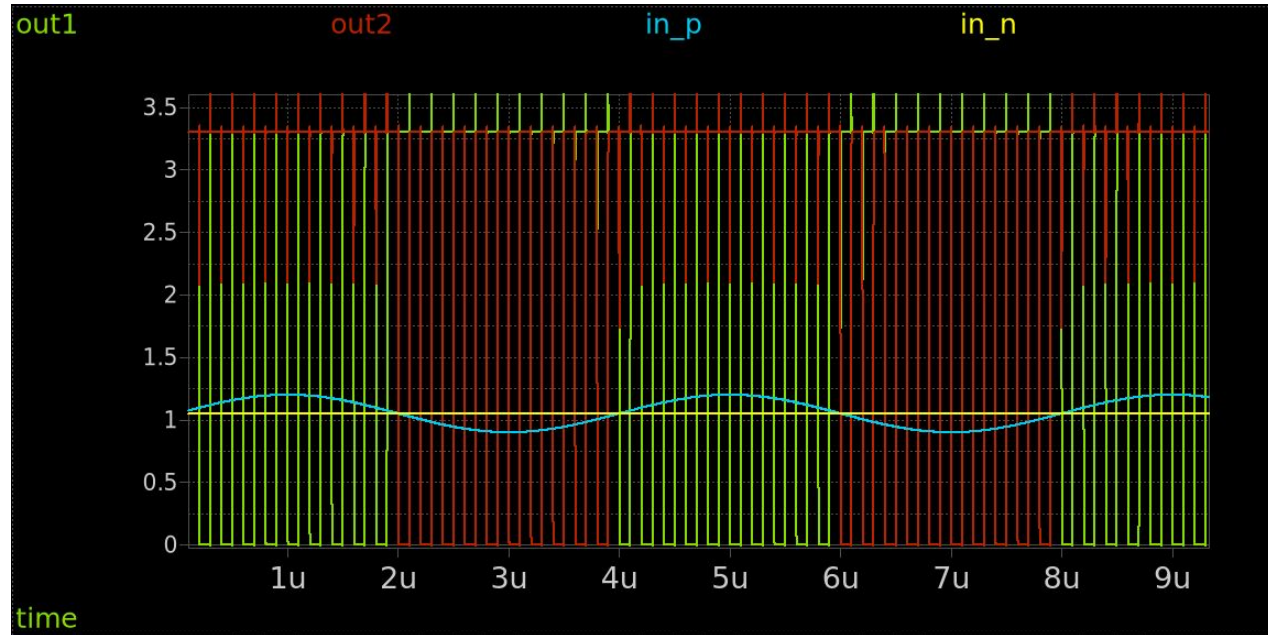
	Units (m)	Units (um)	Pelgrom's Law		
W9	1.50E-05	1.50E+01	$\Delta V_{TH1,2} = \frac{A_{VTH}}{\sqrt{(WL)_{1,2}}}$		
L9	1.00E-06	1.00E+00			
W3	3.00E-06	3.00E+00			
L3	1.00E-06	1.00E+00			
W7	5.00E-06	5.00E+00			
L7	1.00E-06	1.00E+00			
	NMOS	PMOS			
Pelgrom Coefficient (Avth in mV)	6	7	(GF180nm values)		
1st Total Input Variation (mV)	1.549193338	1.807392228			
Pelgrom's Sizing Variation (mV)	2.683281573				
Input Variation from M7 M8 (mV)	1.3	(Found from simulator)			
2nd Total Input Variation (mV)	2.022374842				
Pelgrom's Sizing Variation (mV)	3.464101615				
Input Variation from M3 M4 (mV)	1.2	(Found from simulator)			
3rd Total Input Variation (mV)	2.351595203			Input Capacitance	1.7pF (Output Load of TIA)
			LSB 4.7mV		
If we add preamp gain should be: >	1.00067881		Want < 2.35mV		



- Chose a very large W and L for input nmos transistors to decrease input offset as much as possible without having very high input capacitance would could cause loading issues to TIA
- Added mismatch voltage sources between nodes to find input referred offset voltages

Comparator Full Test

- Used input sine wave from 0.9V to 1.2V as V_{in1}
- Used 1.05V as V_{in2} as the center of the sine wave
- Verified that the comparator works as expected



CDAC

SAR Logic